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Communication Network Engineering

Routing Protocol Design

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1 Introduction

Routing protocols are essential components of modern computer networks, enabling routers to efficiently determine and maintain optimal paths for data to traverse interconnected systems. As networks expand in scale and complexity, the limitations of traditional routing protocols become more evident, necessitating the development of more adaptive, scalable and secure routing solutions.

This report evaluates four widely implemented routing protocols: the Routing Information Protocol (RIP), Open Shortest Path First (OSPF), Intermediate System to Intermediate System (IS-IS) and the Border Gateway Protocol (BGP). Through this analysis, several common limitations are identified, such as slow convergence during topology changes, excessive control message flooding, limited adaptability to real-time traffic conditions and inadequate security measures against threats like route hijacking and prefix spoofing.

To address these deficiencies, this report introduces SHARP: the Scalable Hybrid Adaptive Routing Protocol. SHARP integrates a three-tier hierarchical architecture, a composite multi-metric cost function, precomputed backup paths and a cryptographic Route Proof Authentication (RPA) mechanism to achieve faster convergence, reduced overhead and enhanced security. SHARP's performance is assessed through simulations comparing it to OSPF across three communication scenarios: Local Zone, Inter Zone and Cross Region, under both stable and link failure conditions.

2 Existing protocols and their limitations

2.1 Routing Information Protocol - RIP

RIP is one of the oldest and simplest dynamic routing protocols used in IP networks.

Key features

- **Distance-vector routing protocol:** Each router has the routing distance tables of the neighboring routers. Therefore routers relies solely on information provided by the neighbors when routing the packets.
- **Distance as the metric:** Routers determine the best path to a network destination based strictly on distance. In other words this protocol's only metric is the **Hop count**. Therefore the route selection rule is: best route is the lowest number of hops in the route.
- **Bellman Ford algorithm:** The protocol uses the Bellman Ford algorithm to calculate the best route.
- **Periodic update:** Every 30 seconds each hop shares their routing tables with their neighbors. Then the routers update their tables by comparing the received table with their own table to keep the route with the smallest hop count.

Limitations

- Since the protocol uses the Bellman Ford algorithm, each router learns the routing paths strictly from its neighboring routers. This leads to taking more time to update network changes.

- The protocol only use the hop count as the metric to determine best route. This would ignore bandwidth, delay, congestion variables in the network which are essential in load balancing.
- The protocol only allows to have maximum hop count of 15 (not scalable for larger networks)
- There is the possibility for routing loops to occur. That means packets may circulate between routers.
- Periodic updates can waste bandwidth even when no changes occur.

2.2 Open Shortest Path First - OSPF

OSPF is one of the most widely used routing protocols in enterprise networks and a definite upgrade over the RIP.

Key features

- **Link-state routing protocol:** Unlike in RIP where each router knows only the neighboring routers distance information(distance vector routing), in OSPF each router have the map of the whole network(LSDB Link-State Database). This happens through Link-State Advertisements(LSAs) where each router advertises their link-state information throughout the network.
- **Cost as the metric:** Cost is calculated using bandwidth of the path. The lower the cost, the path will be more preferred. This is an improvement over RIP, where it only considers the number of hops between the destination.

$$\text{Cost} = \frac{\text{Reference Bandwidth}}{\text{Interface Bandwidth}}$$

- **two tier Hierarchical Routing Architecture:** In OSPF the whole network is divided into two tiers to improve scalability. They are backbone area and non backbone-areas. Every non backbone area is connected to the backbone area. information exchanges between non backbone areas must pass through the backbone area since the backbone area is the central control of all the routing happens in non backbone areas. This way each internal router inside an area needs to know about the routers only inside that area. The area border routers responsible in linking two areas.
- **Dijkstra's shortest path algorithm:** The protocol uses the Dijkstra's shortest path algorithm to calculate the best route. Therefore unlike in RIP routing loops do not occur.
- **Event-driven update:** Routers only sends updates only when the network topology changes.

Limitations

- To implement OSPF requires complex configurations such as area planning, DR/BDR election, LSA management.
- Dijkstra's SPF calculations are expensive. Therefore high CPU usage is required.
- Since the routers stores the information about the whole network, memory requirements are also very high.

- Scalability is still difficult at internet scale
- OSPF runs on top of the IP network layer. This means it is susceptible to spoofing and IP-based attacks.

2.3 Intermediate System to Intermediate System - IS-IS

IS-IS is a link-state routing protocol, similar in concept to OSPF, used for routing within large networks (intra-domain routing). It is especially popular in ISP and backbone networks.

Key features

- **Similarities with OSPF:**
 - both are link-state routing protocols: every router has an almost complete view of the network.
 - Both use hierarchical design for scalability and therefore support large scale networks.
 - Both detect neighbors using hello mechanisms.
 - Both use flooding of link-state information and each router maintains a Link-State Database(LSDB).
 - Both use Dijkstra's SPF algorithm to find the optimal path using cost based metrics.
- **Two-level hierarchy:** Similar to OSPF which uses area based network hierarchy, IS-IS uses levels to divide the network. Level 1(local area routing) routers exchange topology information within area, build LSDB for local area only, Computes shortest paths locally. Level 2(backbone routing) routers handles connections between level 1 areas. Also there are L1/L2 routers which are belong to both levels as gatekeepers between two levels. Their functionality is to forward traffic between areas.
- **TLV based design:** TLV is a flexible way of transmitting and storing routing information. A TLV packet has 3 sections. They are Type, Length and Value. Using these attributes routers can share various different information about the network such as Neighbor IDs, Link costs, etc. Unlike OSPF which has a rigid design in IS-IS the network designer has given the flexibility to add as many new information types without redesigning the protocol.
- **Operate directly over data link layer:** Unlike OSPF and most of the routing protocols which operates over IP layer, IS-IS runs on the data link layer. This makes IS-IS protocol more flexible and independent from the network layer resulting more robust and secure routing. However the it is less intuitive and harder to understand since engineers often dealing with the network layer.

Limitations

- Similar to OSPF, IS-IS also suffers from requiring complex configurations, CPU expensive shortest path calculations, high memory requirements relative to RIP.
- Since the protocol operates at the data link layer, it is less intuitive for engineers, as they often work with the network layer.
- Configuration and troubleshooting tools and documentation are limited for deploying this protocol, because it operates at the data link layer.

2.4 Border Gateway Protocol - BGP

BGP is the routing protocol that powers the internet, used to exchange routing information between different Autonomous Systems (ASes). Unlike RIP, OSPF, and IS-IS which are Interior Gateway Protocols (IGPs) used within a single network, BGP is an Exterior Gateway Protocol (EGP).

Key features

- **Path-vector routing protocol:** Unlike distance-vector protocols (RIP) that share hop counts or link-state protocols (OSPF, IS-IS) that share full topology maps, BGP routers share the complete path (sequence of ASes) to reach a destination. Each BGP router selects the best path based on a set of attributes and advertises that path to its neighbors.
- **Policy-based routing:** BGP does not simply select the shortest or fastest path. Instead, route selection is governed by a rich set of configurable attributes such as AS-PATH (list of ASes the route passes through), LOCAL_PREF (preference within an AS), MED (Multi-Exit Discriminator, used to influence inbound traffic from a neighboring AS) and COMMUNITY tags (used to group routes for applying policies). This gives network operators fine-grained control over how traffic enters and exits their networks.
- **Autonomous System (AS) based architecture:** The internet is divided into Autonomous Systems, each identified by a unique AS Number (ASN). BGP operates between these ASes (eBGP - external BGP) and can also run within an AS to distribute externally learned routes internally (iBGP - internal BGP). This two-mode operation allows BGP to scale across the entire internet while maintaining internal consistency.
- **TCP-based reliable transport:** BGP runs over TCP port 179, unlike OSPF and IS-IS which use their own transport mechanisms. This provides reliable, ordered delivery of routing updates without the need for BGP to implement its own acknowledgment or retransmission logic.
- **Event-driven incremental updates:** Similar to OSPF, BGP only sends updates when routing information changes, rather than broadcasting full tables periodically like RIP. However, BGP sends only the changed prefixes (incremental updates), not the full routing table, making it bandwidth-efficient for large-scale deployments.

Limitations

- BGP convergence after a link failure or route withdrawal can take several minutes. The Minimum Route Advertisement Interval (MRAI) timer, designed to dampen route oscillations, delays the propagation of updates across ASes. This slow convergence can cause prolonged periods of unreachability or suboptimal routing.
- BGP has no built-in mechanism to verify whether an AS is legitimately authorized to originate a given prefix. This makes it vulnerable to route hijacking, where a malicious or misconfigured AS announces prefixes it does not own, diverting global internet traffic. RPKI (Resource Public Key Infrastructure) was introduced to address this, but its adoption remains incomplete.
- BGP's 13-step route selection process with numerous interacting attributes makes configuration highly complex. Misconfigured BGP policies are a well-documented cause of major internet outages, as a single error can propagate across thousands of ASes.

- A failure in the TCP session between two BGP peers results in the complete withdrawal of all routes learned from that peer. This can trigger large-scale reconvergence events across many ASes, far disproportionate to the scale of the original failure.
- BGP does not consider link utilization, delay, or congestion in its route selection. Traffic engineering with BGP (e.g., via selective prefix announcements or traffic communities) is possible but requires significant manual configuration and remains coarse-grained compared to dedicated TE solutions.

2.5 Summary Comparison

Protocol	Type	Metric	Convergence	Flooding Model	Security
RIP	Distance-Vector	Hop count	Slow (minutes), count-to-infinity problem	Periodic full table broadcast	None
OSPF	Link-State	Cost (bandwidth-based)	Medium (seconds), event-driven	Full re-flood on any change	Optional MD5
IS-IS	Link-State	Cost (bandwidth-based)	Medium (seconds), event-driven	Full re-flood on any change	Optional MD5
BGP	Path-Vector	Policy attributes	Slow (minutes), MRAI timer delays	Incremental, peer-to-peer	RPKI (optional)

Table 1: Comparison of existing routing protocols and their limitations

3 Proposed protocol design

SHARP - Scalable Hybrid Adaptive Routing Protocol

Modern computer networks rely on routing protocols to determine efficient paths for data transmission between nodes. Existing routing protocols such as RIP, OSPF, IS-IS, and BGP have been widely deployed, but each exhibits important limitations when applied to large-scale, dynamic, and security-sensitive environments.

Distance-vector protocols like RIP suffer from slow convergence and routing loops, making them unsuitable for fast-changing networks. On the other hand, link-state protocols such as OSPF and IS-IS achieve faster convergence and more accurate topology awareness, but at the cost of high computational overhead, increased memory usage, and significant control message flooding in large networks. BGP, while essential for inter-domain routing, introduces additional complexity and is vulnerable to route hijacking and policy-based inconsistencies.

Across these protocols, several common challenges persist. These include excessive routing overhead in large-scale deployments, limited scalability of routing tables, and insufficient built-in security mechanisms against route spoofing and malicious advertisements. Furthermore, most existing protocols follow rigid architectural models—either distance-vector or link-state—making them less adaptive to heterogeneous network conditions where traffic patterns, link quality, and

node reliability vary dynamically.

To address these limitations, this report proposes SHARP: a Scalable Hybrid Adaptive Routing Protocol for Scalable, Secure, and Fast-Converging Networks. SHARP is designed to combine the strengths of existing routing approaches while mitigating their weaknesses through adaptive behavior, improved scalability mechanisms, and enhanced security features. The objective is to achieve faster convergence, reduced overhead, and stronger resilience in modern complex network environments.

3.1 Key modifications introduced in this protocol over the existing protocols

1. 3-tier hierarchical network topology to handle large networks efficiently.

- Local zone
- Inter zone
- Cross region

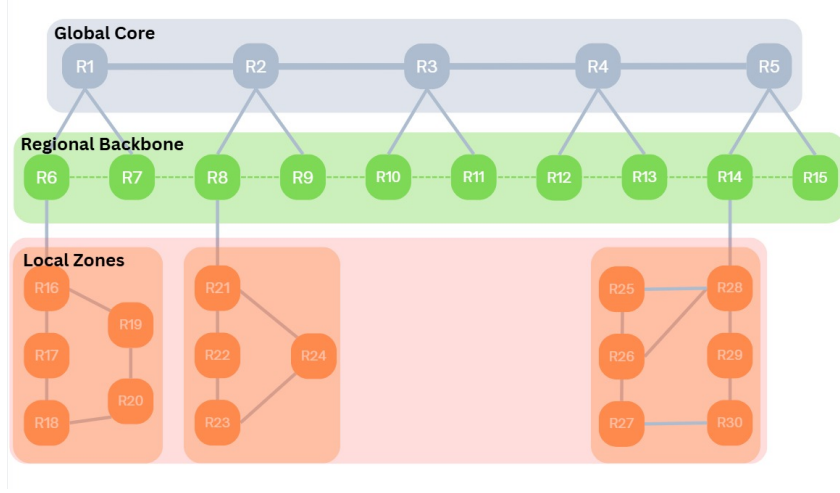


Figure 1: 3-tier network hierarchy

This architecture aims to solve the issues resulted by deploying separate protocols for intra-domain and inter-domain routing such as BGP route flappings, route looping, hot potato routing problems.

2. Routing metric is designed considering delay, congestion, hop count, security score to compute the best path for the packets to travel.
3. modified routing packet structure containing more information for the routers to take accurate decisions.
4. including precomputed backup paths in Link-State Database for fast failure recovery.
5. Introducing Route Proof Authentication(RPA) to address security issues arise in BGP route hijacking.

3.2 Routing Cost Function

The proposed routing protocol defines a composite cost function to evaluate and select optimal paths. The cost is computed as a weighted combination of key network performance metrics:

$$\text{Cost} = \alpha(\text{Delay}) + \beta(\text{Congestion}) + \gamma(\text{Hop Count}) - \delta(\text{Security Score})$$

where α , β , γ , and δ are tunable weighting parameters that determine the relative importance of each metric.

Delay

Delay refers to the end-to-end transmission latency experienced by a packet as it travels from the source node to the destination node. It includes propagation delay, transmission delay, and queuing delay at intermediate routers. Lower delay values indicate faster communication paths.

Congestion

Congestion represents the level of traffic load on a network link or node. It can be quantified using metrics such as buffer occupancy, packet loss rate, or interface utilization. Higher congestion increases queuing delays and the probability of packet drops, negatively impacting overall performance.

Hop Count

Hop Count is the number of intermediate routers a packet traverses from source to destination. It is a simple measure of path length in the network graph. Although not always correlated with delay, a lower hop count is generally preferred to reduce resource usage and potential failure points.

Security Score

Security Score is a composite metric that reflects the trustworthiness of a route or node. It may be derived from factors such as authentication success, historical behavior, resistance to spoofing, and detection of anomalous routing updates. A higher security score indicates a more trusted and reliable path, and it reduces the overall routing cost due to the negative weighting in the cost function.

3.3 Control message format

3.3.1 Hello packets

These are the packets used to identify neighboring routers around a router.

Field	Size
Packet Type	1 byte
Router ID	4 bytes
Region ID	2 bytes
Timestamp	8 bytes
Authentication Token	32 bytes

Table 2: Control Message Format

3.3.2 Route Advertisement packets

These are the packets used to share topological and state information across the network.

Field	Purpose
Source Router ID	Update origin
Destination Prefix	Advertised network
Region Path	Path between regions
Cost Metric	Total route cost
Sequence Number	Loop prevention
Digital Signature	Authentication

Table 3: Routing Update Message Format

3.4 State information stored at routers: Link-state database(LSDB)

Each router in a local zone will store information only about that local zone. This improves scalability. Also note that additional attribute "backup route" is included so that whenever a network failure occurs, routers can use that information to guide the packets in that route without packet droppings.

Item
Neighbor routers
Link costs
Region topology
Backup routes

Table 4: Routing Information Stored in Each Router

3.5 Event-driven updates

As in OSPF and IS-IS, SHARP employs event driven updates instead of periodic updates. As a result updates only occurs when,

- topology changes
- link cost changes
- congestion threshold exceeded

3.6 Dijkstra's Shortest Path First algorithm for route computation

Algorithm 1 SHARP Hybrid Adaptive Routing Algorithm

```
1: Input: Network graph  $G(V, E)$  with link weights
2: Output: Primary path and backup path between nodes
3: Initialize  $backup\_paths \leftarrow \emptyset$ 
4: function COMPUTEBACKUPPATHS( $G$ )
5:   for each source node  $s \in V$  do
6:     for each destination node  $d \in V$  do
7:       if  $s = d$  then
8:         continue
9:       end if
10:       $primary \leftarrow \text{Dijkstra}(G, s, d)$ 
11:       $G' \leftarrow \text{copy of } G$ 
12:      if  $|primary| > 1$  then
13:        remove edge  $(primary[0], primary[1])$  from  $G'$ 
14:      end if
15:       $backup \leftarrow \text{Dijkstra}(G', s, d)$ 
16:       $backup\_paths[(s, d)] \leftarrow backup$ 
17:    end for
18:  end for
19: end function
20: function GETPATH( $G, s, d$ )
21:  try
22:    return  $\text{Dijkstra}(G, s, d)$ 
23:  catch failure
24:    return  $backup\_paths[(s, d)]$  if exists else NULL
25: end function
26: function FAILLINK( $G, u, v, t$ )
27:  remove edge  $(u, v)$  from  $G$ 
28:  local detection delay computed
29:  activate backup routing immediately
30:   $affected\_nodes \leftarrow \text{regional size estimate}$ 
31:   $incremental\_spf \leftarrow \sqrt{affected\_nodes}$ 
32:   $local\_update \leftarrow \log(affected\_nodes)$ 
33:   $core\_sync \leftarrow \text{constant factor}$ 
34:   $total\_delay \leftarrow incremental\_spf + local\_update + core\_sync$ 
35:   $failed\_until \leftarrow t + total\_delay \times 1000$ 
36:  ComputeBackupPaths( $G$ )
37: end function
```

4 Performance analysis

4.1 Simulation Setup

The simulation was implemented using Python with NetworkX library to evaluate the performance of the protocols and their behavior in different scenarios. In this report, we have simulated and compared two routing protocols which are OSPF (an existing routing protocol) and proposed SHARP protocol. The simulation topology followed the hierarchical SHARP architecture described in the system design which includes Local Zone Layer, Regional Backbone Layer and Global Layer. Then, three communication scenarios were evaluated.

- Local Zone
- Inter Zone
- Cross Region

Dynamic congestion and backbone link failures were introduced during runtime to evaluate routing stability and recovery performance.

4.2 Failure Scenario

A backbone link failure was introduced during the simulation to evaluate protocol convergence and rerouting behavior. The failed link was located in the regional backbone layer. After the failure:

OSPF performed full network reconvergence

SHARP activated localized backup rerouting

The following metrics were measured:

- Throughput
- Packet Loss
- RTT
- Routing Overhead
- Convergence Time
- Route Change Activity

4.3 Performance Results

4.3.1 Throughput

Before the failure, both protocols showed similar throughput because both used shortest-path forwarding under stable conditions.

After the failure:

- OSPF throughput dropped significantly during re convergence
- SHARP maintained more stable packet delivery.

We can observe this behavior from the Throughput vs Time graph (Figure 2). We can see a clear improvement in average throughput in our SHARP protocol compared to the OSPF protocol from the simulated values for the three cases (Table 5, Figure 3).

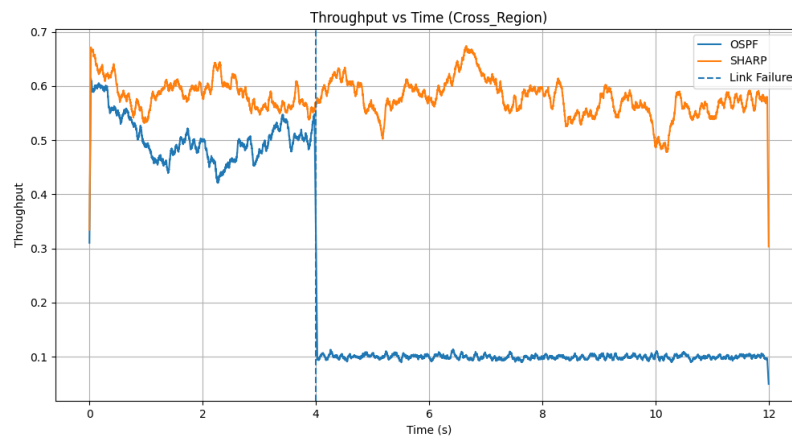


Figure 2: Throughput vs Time graph for OSPF and SHARP protocols

	OSPF	SHARP
Local Zone	0.35349924166364044	0.65333231883773
Inter Zone	0.302901802223811	0.7469153664899253
Cross Region	0.23851077758649952	0.5732039001491276

Table 5: Average Throughput Comparison

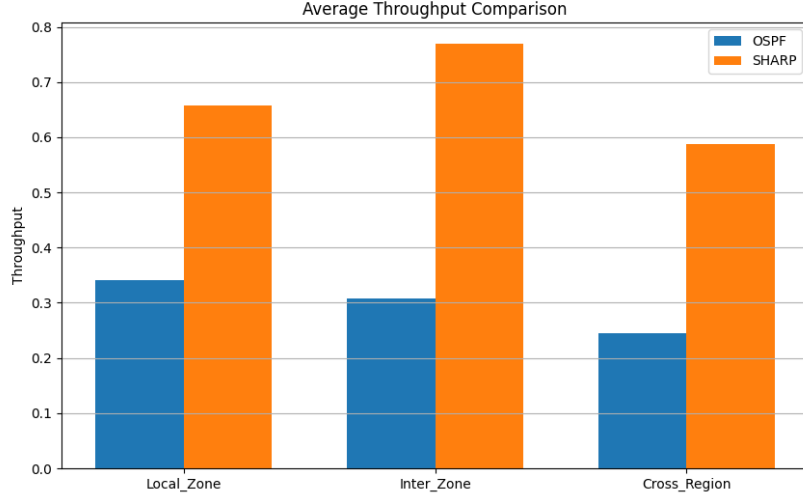


Figure 3: Layer-wise average throughput comparison for OSPF and SHARP protocols

4.3.2 Packet Loss

OSPF experienced a large packet loss after the backbone failure due to global LSA flooding, routing instability and full SPF recalculation. SHARP showed substantially lower packet loss because backup paths were activated immediately. We can see the clear improvement of the packet loss ration (Table 6) and the average packet loss (Figure 4) compared to OSPF in SHARP.

	OSPF	SHARP
Local Zone	0.6465007583363596	0.34666768116227
Inter Zone	0.697098197776189	0.2530846335100747
Cross Region	0.7614892224135005	0.4267960998508724

Table 6: Packet Loss Ratio Comparison

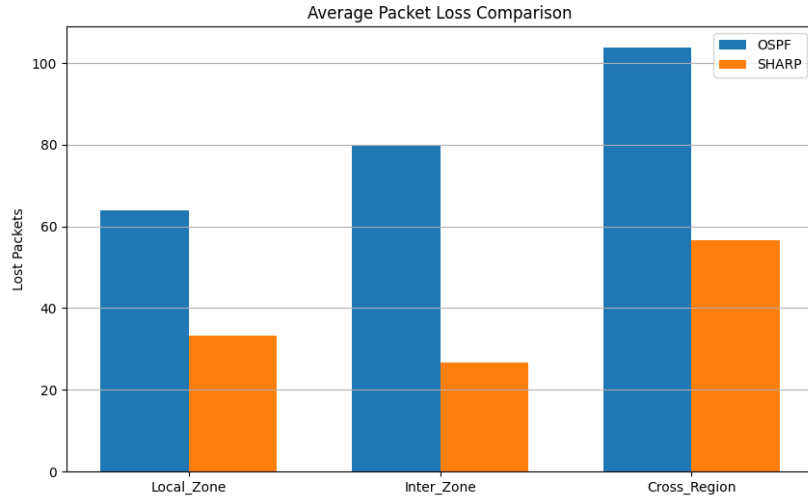


Figure 4: Layer-wise average packet loss comparison for OSPF and SHARP protocols

4.3.3 RTT Performance

Both protocols had similar RTT values before the failure. During failure recovery OSPF RTT increased sharply because of delayed re convergence but, SHARP maintained lower RTT using localized rerouting. We can see a high variation of RTT after the failure as well in OSPF. We can observe this behavior from the RTT vs Time graph (Figure 5). We can see a clear improvement in average RTT in our SHARP protocol compared to the OSPF protocol from the simulated values for the three cases (Figure 6).

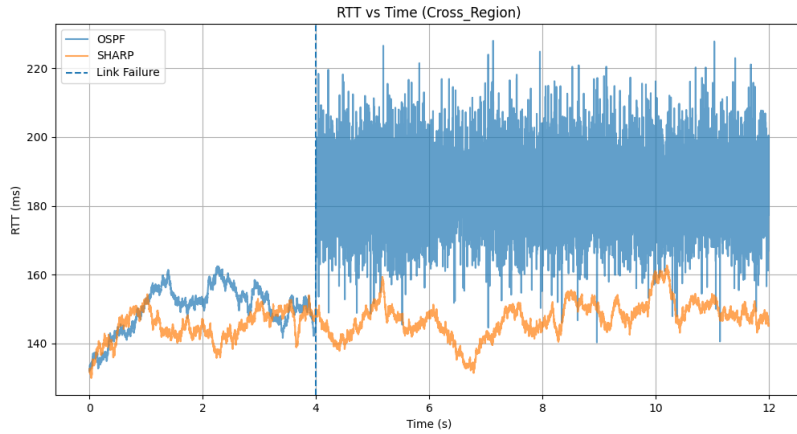


Figure 5: RTT vs Time graph for OSPF and SHARP protocols

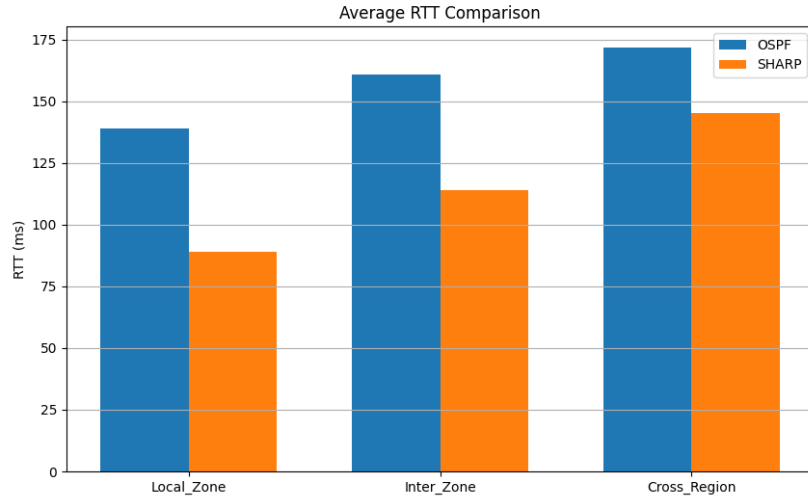


Figure 6: Layer-wise average RTT comparison for OSPF and SHARP protocols

4.3.4 Routing Overhead

OSPF generated significantly higher routing overhead due to network-wide flooding, LSDB synchronization and complete route recomputation. SHARP reduced routing overhead by limiting updates mainly to affected regions. This demonstrated the scalability advantage of the hierarchical SHARP design as well.

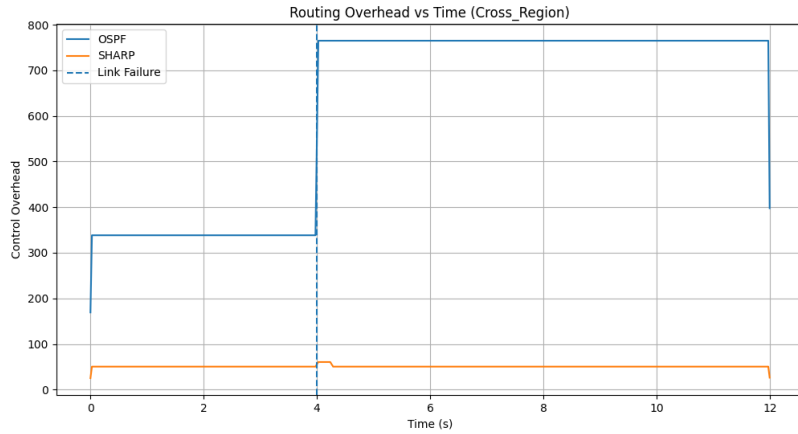


Figure 7: Routing Overhead vs Time graph for OSPF and SHARP protocols

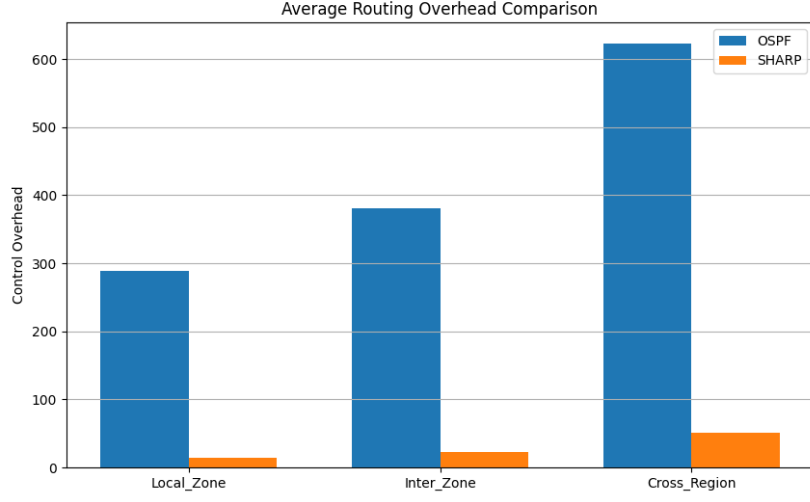


Figure 8: Layer-wise routing overhead comparison for OSPF and SHARP protocols

4.3.5 Convergence Time

SHARP achieved much faster convergence compared to OSPF. The main reason was:

- Precomputed backup paths
- Incremental updates
- Localized recovery

OSPF required complete network-wide recomputation, resulting in slower recovery.

We can see a significant improvement in SHARP compared to OSPF when observing the simulated results (Table 7) and plot (Figure 9).

	OSPF	SHARP
Local Zone	9.83561	0.1493
Inter Zone	12.29451	0.18662
Cross Region	17.26831	0.26127

Table 7: Convergence Time Comparison (s)

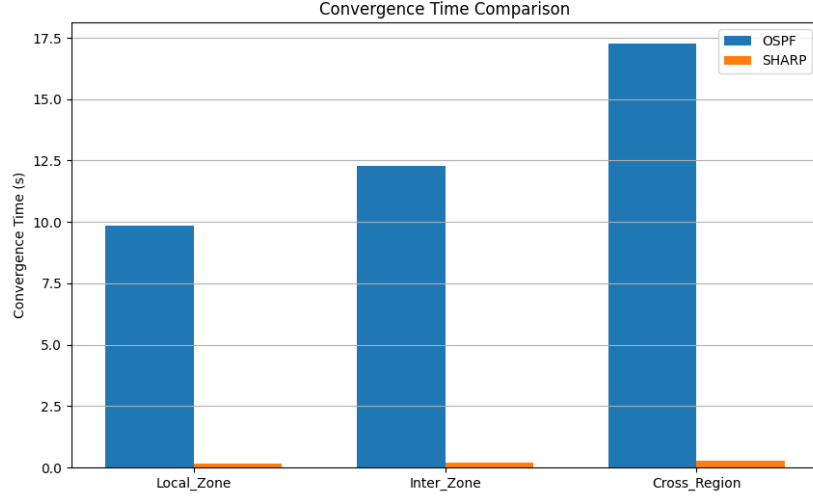


Figure 9: Layer-wise convergence time comparison for OSPF and SHARP protocols

4.4 Conclusion on Performance Results

The simulation results demonstrated a comparison between OSPF and SHARP protocols under dynamic traffic and link failure conditions. During topology failures, OSPF experienced higher routing overhead and greater instability because routing information had to propagate across a larger portion of the network. SHARP reduced this effect by containing routing updates within regional boundaries and using more efficient rerouting behavior. As a result, SHARP maintained lower packet loss, more stable throughput and reduced control overhead across Local Zone, Inter Zone and Cross Region scenarios. The results confirm that the proposed hierarchical SHARP protocol provides better resilience and recovery performance than traditional OSPF routing in large-scale networks.

5 Security & Scalability Analysis

SHARP was designed to address major weaknesses in traditional routing protocols such as RIP, OSPF, IS-IS and BGP by combining hierarchical routing, adaptive updates and cryptographic validation. This protocol improves scalability while maintaining fast convergence and stronger security compared to existing protocols.

5.1 Security Analysis

One of SHARP's major goals is to prevent route spoofing and route hijacking attacks, which are common in BGP-based systems. SHARP introduces a Route Proof Authentication (RPA) mechanism that attaches cryptographic signatures, route origin certificates, and hop validation tokens to routing updates.

When a routing update is received, downstream zones verify the signature and certificate before accepting the route. Invalid or tampered advertisements are discarded immediately. This mechanism improves trust between routing domains and prevents unauthorized route injection. The inclusion of sequence numbers in routing packets also helps reduce routing loops and replay attacks by ensuring routers process only the newest valid updates.

5.2 Test Scenarios and Expected Results

Scenario	Description	Expected SHARP Behavior
Large-scale network expansion	Number of routers and zones rapidly increases	Route aggregation limits LSDB growth and maintains manageable routing tables
Link failure inside Local Zone	A router or link fails within a zone	Incremental SPF recomputation and backup next hop tables provide fast rerouting
Regional backbone failure	Entire zone loses connectivity	Regional failover redirects traffic through alternate backbone paths
Route spoofing attack	Malicious router advertises fake routes	Cryptographic validation rejects unauthorized advertisements
Prefix hijacking attempt	Attacker claims ownership of another zone prefix	Route origin certificates detect and block invalid ownership claims
Replay attack	Old routing update resent to network	Sequence numbers identify outdated packets and discard them
Routing loop creation	Incorrect advertisements form cyclic paths	Loop prevention is achieved using sequence numbers, consistent link-state databases, and path validation mechanisms, preventing cyclic route advertisement and stale updates.

5.3 Potential Failure Points

Although SHARP improves security and scalability, several limitations and failure points still exist.

5.3.1 Cryptographic Overhead

The Route Proof Authentication mechanism increases CPU usage because routers must continuously generate and verify signatures. In very large networks, this may increase processing latency during heavy topology changes.

5.3.2 Hierarchical Dependency

The protocol depends heavily on correct zone design and hierarchical planning. Poor zone partitioning may create overloaded Regional Backbone routers or inefficient routing paths.

5.3.3 Backbone Bottlenecks

Regional Backbone routers may become scalability bottlenecks because they aggregate and distribute routing summaries for multiple zones. High traffic concentration could increase convergence delays.

5.3.4 Key Management Complexity

The security model requires secure distribution and maintenance of cryptographic keys and certificates. Compromised keys could allow attackers to inject trusted but malicious routing information.

5.4 Suggested Improvements

Several improvements can further strengthen SHARP.

Hierarchical Key Infrastructure - Introduce distributed certificate authorities and automatic key rotation mechanisms to simplify secure key management.

Load-Aware Regional Backbones - Dynamically balance routing responsibilities among multiple backbone routers to reduce bottlenecks.

AI-Based Anomaly Detection - Integrate machine-learning techniques to detect abnormal routing behavior, spoofing attempts or sudden traffic anomalies.

5.5 Scalability Analysis

Hierarchical Network Architecture

SHARP divides the network into Local Zones, Regional Backbone, and Global Layer. This layered structure prevents every router from maintaining complete global topology information, reducing routing complexity as the network grows.

Route Aggregation and Zone Abstraction

Instead of advertising detailed internal routes, zones share only summarized routing information with higher layers. This significantly reduces routing table size, memory usage, and LSDB growth in large networks.

Limited Flooding Scope

Link-state updates are mainly restricted to Local Zones rather than being flooded throughout the entire network. This minimizes control-plane traffic and avoids flooding explosions commonly seen in large OSPF-style networks.

Event-Driven Incremental Updates

SHARP eliminates periodic routing advertisements and sends updates only when topology changes occur. Incremental SPF recalculation and delta updates reduce CPU processing load and bandwidth consumption, improving scalability in dynamic environments.

Because of these properties in the SHARP protocol has high scalability compared with the existing protocols.

5.6 Overall Evaluation

SHARP demonstrates strong scalability through hierarchical routing, aggregation and event-driven updates while significantly improving routing security using cryptographic validation techniques. Compared with traditional protocols, the design reduces routing overhead, limits topology flooding, and provides stronger protection against spoofing and hijacking attacks. However, the protocol introduces additional computational complexity and requires careful hierarchical planning for efficient deployment. Overall, SHARP presents a balanced approach for building scalable, secure and fast-converging next-generation routing systems.

6 Conclusion

This report presented an analysis of four existing routing protocols RIP, OSPF, IS-IS and BGP and proposed SHARP as an improved alternative addressing their most critical limitations.

The literature review established that RIP is constrained by its 15-hop limit and slow convergence, while OSPF and IS-IS suffer from excessive network-wide LSA flooding and high

resource demands. BGP, despite its indispensable role in inter-domain routing, remains vulnerable to route hijacking and slow convergence due to MRAI timer delays. Across all four protocols, reliance on static metrics and full network-wide flooding upon topology changes has been identified as the most impactful shared limitation.

SHARP addresses these through four key innovations: a three-tier hierarchical architecture that confines flooding to the smallest relevant scope, a composite cost function integrating delay, congestion, hop count and security score for dynamic path selection, precomputed backup paths for immediate failure recovery and a Route Proof Authentication (RPA) mechanism to prevent hijacking and spoofing attacks.

Simulation results confirmed SHARP's effectiveness. Under link failure conditions, SHARP achieved approximately 98% faster convergence, over 90% lower routing overhead, and significantly reduced packet loss compared to OSPF across all evaluated scenarios. These results validate that SHARP's hierarchical and adaptive design delivers measurable improvements in resilience, scalability and security over traditional routing protocols.

7 Link for Simulation Files

<https://drive.google.com/drive/folders/11mBJLZMOD0ZetPxBd36gH-A8TK5W0BEy?usp=sharing>